

Topic 1.4 - Polynomial Functions and Rates of Change

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

Higher-degree polynomials model changing rates more flexibly than linear or quadratic functions. Their formulas, tables, and graphs must be read together: finite differences suggest degree, rates describe intervals, technology locates features, and context restricts where a model is meaningful.

Learning targets

- Use finite differences and polynomial structure to identify or justify a likely degree.
- Analyze polynomial rates, turning behavior, zeros, and representations without overclaiming from secant evidence.
- Use technology responsibly to locate features and critique a polynomial model's domain.

Language and structure

Term	Meaning in this topic
polynomial degree	The greatest exponent with nonzero coefficient.
finite-difference order	The level at which equal-step differences become constant.
turning point	A location where a graph changes between increasing and decreasing.
local rate estimate	A short-interval secant rate near an input.
model domain	The interval on which a formula is intended to represent a context.

Launch - reason before calculating

For $x = 0, 1, 2, 3, 4$, outputs are 1, 2, 9, 28, 65. Build successive difference rows, identify a likely degree, and explain why a contextual model need not remain valid for all real inputs.

Concept 1: Degree and finite differences

Core idea

For exact polynomial values at equally spaced inputs, a degree- n polynomial has constant n th differences and zero $(n + 1)$ st differences. This gives the least likely degree for exact data; measured data usually show only approximate patterns.

Worked example - annotate the reasoning

Problem. For $x = 0, 1, 2, 3, 4$, values are 2, 0, -2, 2, 18. Find the least polynomial degree.

Model reasoning. First differences -2, -2, 4, 16; second 0, 6, 12; third 6, 6. Least degree 3.

Check your understanding

Independent

No calculator

For outputs 3, 5, 19, 57, 131 at unit-spaced inputs, determine the least degree.

Concept 2: Rates, direction, and turning behavior

Core idea

Adjacent average rates show net increase or decrease. Changing rate signs can suggest turning behavior, but exact turning inputs may lie inside intervals. Equal endpoint values give average rate zero but do not imply constancy. A degree- n polynomial has at most $n - 1$ turning points.

Worked example - annotate the reasoning

Problem. For $P(x) = x^3 - 6x^2 + 9x + 2$, find one-unit rates from $x = 0$ through 4 and describe displayed direction.

Model reasoning. Values 2, 6, 4, 2, 6 give rates 4, -2, -2, 4: increase, then decrease, then increase, suggesting two turns.

Check your understanding

Independent

No calculator

For $q(x) = x^4 - 4x^2$, use values at -2, -1, 0, 1, 2 to describe direction.

Concept 3: Technology, local rates, and model limits

Core idea

A graphing utility can locate approximate zeros and extrema and estimate a local rate with a short interval. Evidence should include a window, domain, and precision. Polynomial end behavior can create impossible outputs outside a contextual interval.

Worked example - annotate the reasoning

Problem. For $F(x) = 0.5x^4 - 4x^2 + 3$, use $F(1.9) \approx -4.92395$ and $F(2.1) \approx -4.91595$ to estimate the rate near 2.

Model reasoning. The symmetric rate is $0.008/0.2 = 0.04$, close to zero, suggesting a nearly horizontal graph near 2 without proving an exact rate.

Check your understanding

Independent

Calculator permitted

A calculator gives $G(3.99) = 10.202$, $G(4.01) = 10.198$. Estimate the rate near 4.

Representation laboratory

Connect at least two representations

A polynomial has $f(-2) = 0$, $f(-1) = -6$, $f(0) = 0$, $f(1) = 6$, $f(2) = 0$. Describe signs and consecutive rates, then propose and verify a factored polynomial.

Challenge laboratory

Challenge - justify every claim

Construct a cubic with zeros $-2, 1, 4$ and $P(0) = 16$. Find its average rate on $[0, 2]$ and explain what the result does and does not communicate.

Error clinic and calculator strategy

Common errors to diagnose

- Applying finite differences to unequally spaced inputs.
- Claiming a degree- n polynomial must have exactly n real zeros or $n - 1$ turns.
- Treating an average rate of zero as proof of a constant function.
- Trusting a calculator graph without checking window and contextual domain.

Technology decision

Save the model equation, window, and relevant coordinates. Verify approximate features by substitution or sign checks. Residuals and context matter more than an impressive fit statistic, especially for high-degree extrapolation.

AP-style synthesis**Multi-part reasoning task**

A polynomial height model $H(t)$ is used on $0 \leq t \leq 10$. A graphing utility reports local extrema at approximately $(0.847, 30.319)$, $(4.665, 20.333)$, $(9.488, 38.973)$. Also $H(0) = 28$, $H(10) = 38$.

A. Describe increasing and decreasing intervals suggested by the extrema.

B. Determine absolute maximum and minimum on the contextual domain.

C. Explain why behavior for $t > 10$ should not automatically be interpreted as height.

Exit ticket**Before you leave**

1. Which finite differences are constant for exact cubic data?

2. Maximum turning points for degree 7?

3. Why can an algebraically valid polynomial be contextually invalid?
